ALGEBRAICITY IS STABLE IN TOWERS OF FIELD EXTENSIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The author highlights after the lemma (source lines 978-981) that this descent-to-finitely-generated-subextension idea recurs throughout algebra; the remark is outside the counted span.

Editorial note. The label lemma-algebraic-permanence names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Same chapter: lemma-algebraic-finitely-generated (an algebraic finitely generated extension is finite), lemma-multiplicativity-degrees (degrees multiply in towers, delivered card L48338)

Source reference. Original reference 09GH: sources/stacks-fields/pinned-original-source.tex, source lines 910–910. Prerequisite pointer invoked at 972; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09G9: sources/stacks-fields/pinned-original-source.tex, source lines 759–759. Prerequisite pointer invoked at 974; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let E/k and F/E be algebraic extensions of fields. Then F/k is an algebraic extension of fields.*

Proof. Choose $\alpha \in F$. Then α is algebraic over E. The key observation is that α is algebraic over a finitely generated subextension of k. That is, there is a finite set $S \subset E$ such that α is algebraic over $k(S)$: this is clear because being algebraic means that a certain polynomial in $E[x]$ that α satisfies exists, and as S we can take the coefficients of this polynomial. It follows that α is algebraic over $k(S)$. In particular, the extension $k(S, \alpha)/k(S)$ is finite. Since S is a finite set, and $k(S)/k$ is algebraic, Lemma 09GH shows that $k(S)/k$ is finite. Using multiplicativity (Lemma 09G9) we find that $k(S, \alpha)/k$ is finite, so α is algebraic over k. □

Date: stacks-project:fields.tex:local-snapshot-20261012; GNU FDL 1.2 or later.